

Engineering Mathematics Study Notes: Raabe's Test with Variable Terms

1. IDENTIFY

The mathematical concept presented on the board is **Raabe's Test** applied to an infinite series containing a variable parameter x . When a variable like x is present, **D'Alembert's Ratio Test** is always performed first to determine convergence conditions dependent on x . Raabe's Test is then applied specifically to resolve the boundary case where the ratio test fails (i.e., when the expression involving x equals 1).

2. THE FORMULA

The general term of the series contains a variable $x > 0$. The evaluation follows a two-tiered formula approach:

Tier 1: D'Alembert's Ratio Test

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = L$$

- If $L < 1$, the series converges.
- If $L > 1$, the series diverges.
- If $L = 1$, the test fails (this occurs at a specific value of x).

Tier 2: Raabe's Test Fallback

When $L = 1$, substitute the failing value of x back into the terms and evaluate:

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = l$$

- Convergent if $l > 1$.
 - Divergent if $l < 1$.
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3. THE ALGORITHM

Step 1: Extract the General Term

Define u_n from the given summation notation.

Step 2: Extract the Successor Term

Find u_{n+1} by changing every n to $(n + 1)$. Remember to expand factorials and product chains to include their next natural terms.

Step 3: Run the Ratio Test

Set up $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n}$ and simplify it by canceling common terms. This will leave an expression containing x .

Step 4: State Conditional Convergence

Set the simplified limit expression < 1 for convergence and > 1 for divergence. Isolate x to find the intervals.

Step 5: Isolate the Failure Point

Set the limit expression exactly equal to 1 to find the specific value of x where D'Alembert's test fails.

Step 6: Execute Raabe's Test at the Failure Point

Substitute this specific value of x into your simplified expression for $\frac{u_n}{u_{n+1}}$, plug it into the Raabe formula, compute the limit as $n \rightarrow \infty$, and apply the decision rules.

4. THE MECHANICS

Core Operations Explained

- **Factorial Expansion:** The term $n!$ represents $1 \cdot 2 \cdot 3 \dots n$. Therefore, its successor $(n+1)!$ can be written as $n! \cdot (n+1)$. This allows $n!$ to be completely canceled during division.
- **Product Chain Expansion:** The numerator chain grows by adding 3 to each successive term (4, 7, 10...). The last term is $3n+1$. The next term after it is obtained by replacing n with $n+1$:

$$3(n+1) + 1 = 3n + 3 + 1 = 3n + 4$$

- **Variable Power Split:** The term x^{n+1} can be split algebraically using exponent laws:

$$x^{n+1} = x^n \cdot x^1$$

This allows x^n to cancel out, leaving a single x in the ratio.

Full Mechanics Walkthrough: Problem Solving

Consider the series:

$$\sum \frac{4 \cdot 7 \dots (3n+1)x^n}{n!}$$

The general term u_n is defined as:

$$u_n = \frac{4 \cdot 7 \dots (3n+1)x^n}{n!}$$

The successor term u_{n+1} is constructed by appending the next elements in the patterns:

$$u_{n+1} = \frac{4 \cdot 7 \dots (3n+1)(3n+4)x^{n+1}}{(n+1)!}$$

Set up the standard ratio test limit expression:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{\frac{4 \cdot 7 \dots (3n+1)(3n+4)x^{n+1}}{(n+1)!}}{\frac{4 \cdot 7 \dots (3n+1)x^n}{n!}}$$

Convert the division into direct multiplication by using the reciprocal:

$$\lim_{n \rightarrow \infty} \frac{4 \cdot 7 \dots (3n+1)(3n+4)x^n \cdot x}{n! \cdot (n+1)} \times \frac{n!}{4 \cdot 7 \dots (3n+1)x^n}$$

Cancel all matching terms ($4 \dots (3n+1)$, $n!$, and x^n) appearing simultaneously across the fractions:

$$\lim_{n \rightarrow \infty} \frac{(3n+4)x}{n+1}$$

Factor out the variable n from both the numerator and denominator structures:

$$\lim_{n \rightarrow \infty} \frac{n \left(3 + \frac{4}{n}\right) x}{n \left(1 + \frac{1}{n}\right)}$$

Cancel the isolated factor n from the ratio expression:

$$\lim_{n \rightarrow \infty} \frac{\left(3 + \frac{4}{n}\right) x}{1 + \frac{1}{n}}$$

Apply the limit condition values as n goes to infinity:

$$\frac{(3 + 0)x}{1 + 0} = 3x$$

Apply D'Alembert's conditional decision rules:

- The series is **convergent** if $3x < 1$, which means $x < \frac{1}{3}$.
- The series is **divergent** if $3x > 1$, which means $x > \frac{1}{3}$.
- The test **fails** if $3x = 1$, which means $x = \frac{1}{3}$.

To resolve the failure point, look back at the ratio of terms. Since $\frac{u_{n+1}}{u_n} = \frac{3n+4}{n+1}x$, substitute $x = \frac{1}{3}$ into it:

$$\frac{u_{n+1}}{u_n} = \frac{3n+4}{n+1} \cdot \frac{1}{3} = \frac{3n+4}{3n+3}$$

Invert this expression to find the required value for Raabe's Test:

$$\frac{u_n}{u_{n+1}} = \frac{3n+3}{3n+4}$$

Initiate the Raabe's Test process using this inverted term:

$$\lim_{n \rightarrow \infty} n \left(\frac{3n+3}{3n+4} - 1 \right)$$

Find a common denominator to combine the components inside the parentheses:

$$\lim_{n \rightarrow \infty} n \left(\frac{(3n+3) - (3n+4)}{3n+4} \right)$$

Simplify the numerator expression by distributing the negative sign:

$$\lim_{n \rightarrow \infty} n \left(\frac{3n+3-3n-4}{3n+4} \right)$$

Combine constants to simplify the remaining expression values:

$$\lim_{n \rightarrow \infty} n \left(\frac{-1}{3n+4} \right)$$

Multiply the external factor n into the fractional expression:

$$\lim_{n \rightarrow \infty} \frac{-n}{3n+4}$$

Factor out n from the terms located in the denominator:

$$\lim_{n \rightarrow \infty} \frac{-n}{n \left(3 + \frac{4}{n}\right)}$$

Cancel out the variable n present across both fields:

$$\lim_{n \rightarrow \infty} \frac{-1}{3 + \frac{4}{n}}$$

Apply the final limit condition boundaries to resolve the constant expression:

$$\frac{-1}{3 + 0} = -\frac{1}{3}$$

Compare the final result against the Raabe decision threshold rule:

$$-\frac{1}{3} < 1$$

Since the computed limit value satisfies $l < 1$, the series is **divergent** at $x = \frac{1}{3}$.

Final Summary of Convergence Conditions

Convergent if: $x < \frac{1}{3}$

Divergent if: $x \geq \frac{1}{3}$